

Assignment #3

Course: **CHE 341. CHEMICAL PROCESS CONTROL**
Term: **Spring 2020**
Professor: **Dr. Michael Caracotsios**
Office: **EIB 244**
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Your name: William Rawson

Due Date: Wednesday February 19 by 5:00pm in my office

Number of points.....100

Your score..... 98

Note: All problems are worth 100 points. Your final score will be the average value

Set Points

The set points for the controllers described in this report were determined by running an Athena mathematical model (using macroscopic mass and energy balances). Conditions were set within the Athena model to meet the given specifications of the CSTR for the esterification of acetic acid with ethanol. A new specification used, compared to the model in report 1, which was a 97 gpm flow rate of cooling water at 25°C with a corresponding 60% valve opening. Due to this new condition, the Athena model provided new steady state data that were used as set points for the dynamic controller models.

Table 1: Set Points for all controllers

Feedback Controller Setpoint	Temperature of the reactor	78.60 °C
Cascade Controller Setpoint	Temperature of jacket	45.59 °C
Feed Forward Controller Setpoint	Temperature of cooling water at the inlet	25.00 °C

Process Gains

Table 2: Feedback controller variables used to calculate the average process gain by manipulating the percent opening of the valve

CO Initial %	CO Final %	$T_{R,initial}$ (°C)	$T_{R,final}$ (°C)	$K_{P,FB}$
60	70	78.95	75.16	-0.379
70	50	75.16	83.87	-0.4355
50	60	83.87	78.95	-0.492
Average $K_{P,FB}$:				-0.436

Table 3: Cascade controller variables used to calculate the average process gain by manipulating the percent opening of the valve

CO Initial %	CO Final %	$T_{Jacket,initial}$ (°C)	$T_{Jacket,final}$ (°C)	$K_{P1,CS}$
60	70	45.59	38.49	-0.71
70	50	38.49	54.78	-0.8145
50	60	54.78	45.59	-0.919
Average $K_{P1,CS}$:				-0.815

Table 4: Cascade controller variables used to calculate the average process gain by manipulating the setpoint of the cooling jacket temperature

$T_{\text{setpoint,initial}} (^{\circ}\text{C})$	$T_{\text{setpoint,final}} (^{\circ}\text{C})$	$T_{r,\text{initial}} (^{\circ}\text{C})$	$T_{r,\text{final}} (^{\circ}\text{C})$	$K_{P2,CS}$
45.59	50.59	78.96	81.63	0.534
50.59	40.59	81.63	76.28	0.535
40.59	45.59	76.28	78.96	0.536
Average $K_{P2,CS}$:				0.535

Table 5: Feed Forward controller variables used to calculate the average process gain by manipulating the temperature of the cooling water inlet

$T_{\text{Cool,in Initial}} (^{\circ}\text{C})$	$T_{\text{Cool,in Final}} (^{\circ}\text{C})$	$T_{r,\text{initial}} (^{\circ}\text{C})$	$T_{r,\text{final}} (^{\circ}\text{C})$	$K_{P,FF}$
25	30	78.95	81.03	0.416
30	20	81.03	76.88	0.415
20	25	76.88	78.95	0.414
Average $K_{P,FF}$:				0.415

Units
-2

Calculations

Volume of the Cooling Jacket:

The volume of the cooling jacket was calculated using the fact that the residence time of the cooling jacket is 10% of the residence time of the reactor. With a given feed flowrate of 120 gallons per minute, the residence time of the reactor and jacket can be solved. Next, the volume of the cooling jacket can be back calculated using the calculated residence time of the jacket.

$$(1) \text{Residence Time Jacket} = \frac{1}{10} * \text{Residence Time Reactor}$$

$$(2) \text{Residence Time Reactor} = \frac{\text{Volume}}{\text{Flowrate}}$$

$$\text{Volume of the Cooling Jacket} = 0.808 \text{ m}^3$$

Pressure drop through the valve:

The differential pressure drop through the 3" valve was calculated to be used in the energy balance for the cooling jacket.

$$(1) Q = F_p * K * C_v(x) * \sqrt{\frac{\Delta P}{S_g}}$$

$Q = 97 \text{ gpm}$, $F_p = 1$ (geometry factor), $K = 1$ (if the flowrate units of Q are in gpm), $C_v(x) = 36$ (@ 60% opening), and $S_g = 1$ (for water)

$$\Delta P = 7.26 \text{ psi}$$

Overall heat transfer coefficient:

$$(1) Q = UA(T_c - T_r) \Rightarrow UA = \frac{Q}{(T_c - T_r)}$$

$$Q = 431,725 \text{ W}, T_r = 70, T_c = 42.09$$

*heat duty Q and the temperature of the cooling water were taken from an ASPEN simulation

$$UA = 15,468 \text{ J/s} \cdot K$$

Equations and Parameters**Table 6: Model Equations and their parameters in Athena**

Component A balance (acetic acid)	$F(1) = V * \frac{dCa}{dt} = Q * Ca_{Feed} - Q * Ca - R1 * V$	$R1 = -kCa * Cb$	$U(1) = Ca$
Component B balance (ethanol)	$F(2) = V * \frac{dCb}{dt} = Q * Cb_{Feed} - Q * Cb - R1 * V$	$R1 = -kCa * Cb$	$U(2) = Cb$
Component C balance (ethyl acetate)	$F(3) = V * \frac{dCc}{dt} = Q * Cc_{Feed} - Q * Cc + R1 * V$	$R2 = kCa * Cb$	$U(3) = Cc$
Component D balance (water)	$F(4) = V * \frac{dCd}{dt} = Q * Cd_{Feed} - Q * Cd + R1 * V$	$R2 = kCa * Cb$	$U(4) = Cd$
Energy Balance For Reactor	$F(5) = \rho * Cp * V * \frac{dT}{dt} = Q * \rho * Cp * (T_{feed} - T) + (-\Delta H) * R1 * V - UA * (T - T_c)$		$U(5) = Temp$
Energy Balance For Cooling Jacket	$F(6) = \rho, water * Cp, water * Cv(x) * \sqrt{\Delta P} * (Tw, in - Tw) - UA(Tw - Tr)$		$U(6) = Tempw$

*Assumptions for both simplified energy balances shown above: The density, Cp, and level (height of the liquid in the tank) are constant. Also, the properties at the outlet of the reactor and the outlet of the jacket are assumed to be the same as inside the reactor and jacket, respectively. This is a result of the assumption for the CSTR that perfect mixing occurs

Process Gain Equations:

- Process Gain for Feedback Controller**

$$K_{P,FB} = \frac{\Delta^\circ C \text{ (Reactor Temperature)}}{\% \Delta(CO)}$$

- Process Gain for Cascade Controller**

$$K_{P1,CS} = \frac{\Delta^\circ C \text{ (Cooling Jacket Temperature)}}{\% \Delta(CO)}$$

$$K_{P2,CS} = \frac{\Delta^{\circ}\text{C (Reactor Temperature)}}{\Delta^{\circ}\text{C (Cascade Set Point Temperature)}}$$

- **Process Gain for Feed Forward Controller**

$$K_{P,FF} = \frac{\Delta^{\circ}\text{C (Reactor Temperature)}}{\Delta^{\circ}\text{C (Cooling Water Inlet Temperature)}}$$

Table 7: Parameters used in Athena and their descriptions

Parameter/ Athena Variable	Value	Description, units
Q	0.365	Feed Volumetric flowrate, m ³ /min
CaFeed	3.217	Molar flowrate for acetic acid, kmol/min
CbFeed	3.117	Molar flowrate for ethanol, kmol/min
CcFeed	0.0	Molar flowrate for ethyl acetate, kmol/min
CdFeed	0.167	Molar flowrate for water, kmol/min
Tfeed	25	Temperature of feed, °C
Vol	0.5	Volume of the reactor, m ³
UA	928.0	Overall heat transfer coefficient, kJ/min*K
Tc	50	Temperature of cooling water, °C
rhoFM	17.789	Molar density of the feed, kmol/m ³
rhoWM	55.18	kJ/min-K
Cpf	100.032	Heat capacity of feed, kJ/kmol*K
DHR1	-29.07*10 ³	Heat of reaction, kJ/kmol
k0fw	1.9*10 ⁸	Pre-exponential factor for forward reaction, m ³ /kmol*s
k0r	5.0*10 ⁷	Pre-exponential factor for reverse reaction, m ³ /kmol*s
Efw	59,500	Activation energy, J/mol
Er	59,500	Activation energy, J/mol
R1	kf * Ca * Cb - kr * Cc * Cd	Reaction rate, mol/L*s
Tempw	45.59	Temperature of jacket, °C
Qcw	97.0	Cooling water flowrate, gpm
Tcin	25.0	Temperature of water inlet, °C
Vjack	0.808	Volume of jacket, m ³
sqrtP	2.694	Square root of pressure drop in 3" valve, psi ^{1/2}

Graphical Results

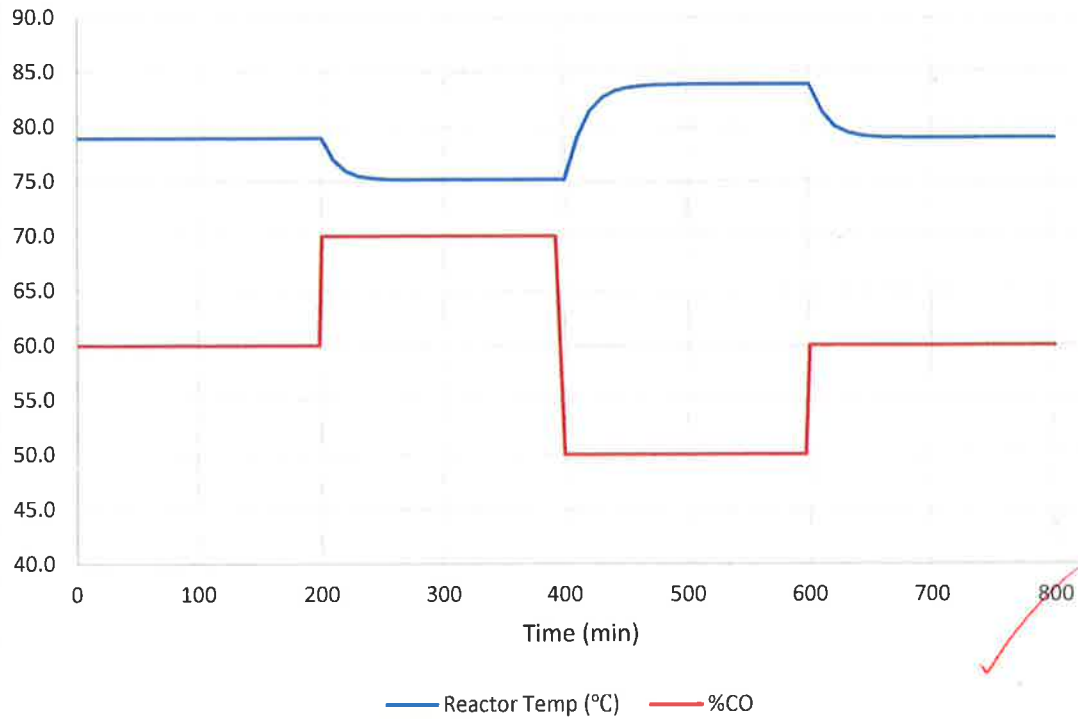


Figure 1: Feedback controller response from the change in %(CO) and reactor temperature versus time

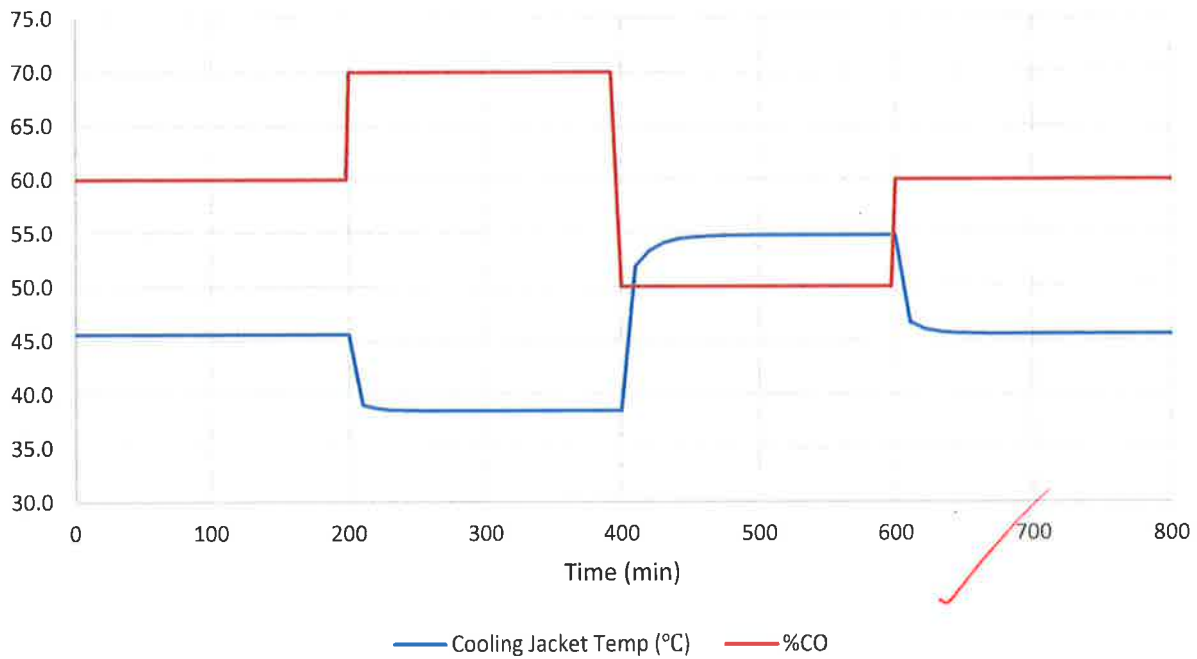


Figure 2: Cascade controller response from the change in %(CO) and cooling jacket temperature versus time

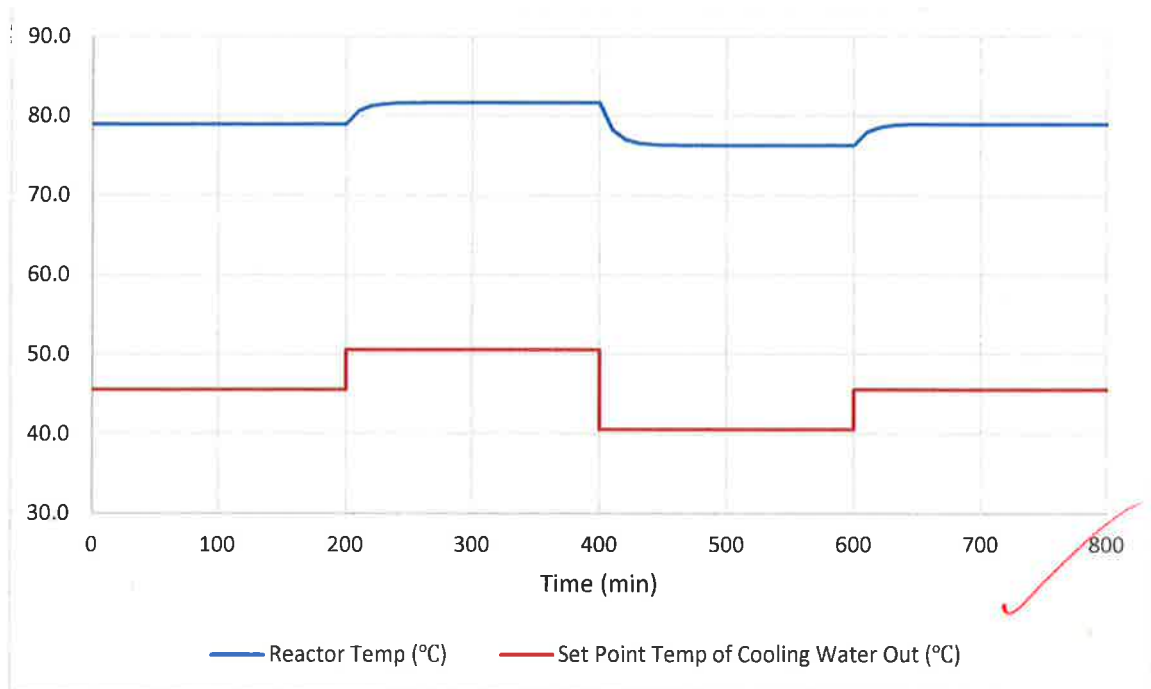


Figure 3: Cascade controller response from the change in set point (T_w) and reactor temperature versus time

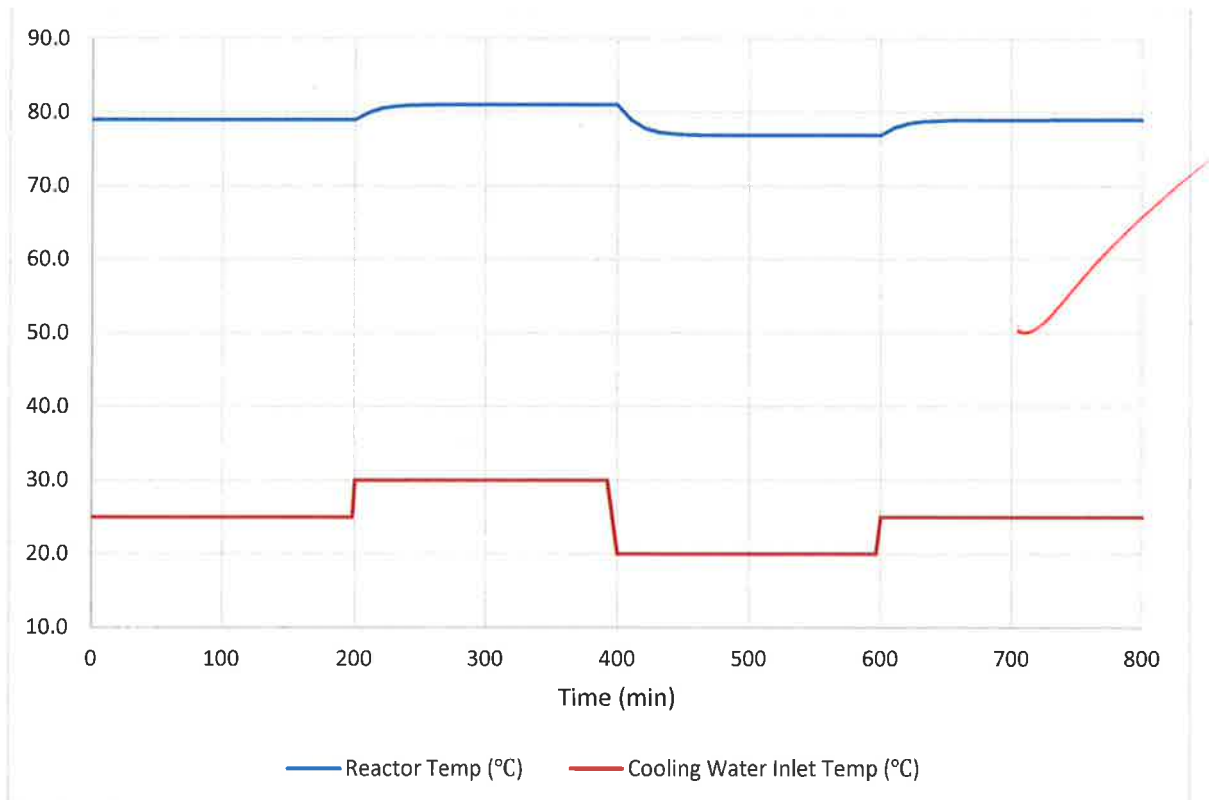


Figure 4: Feed forward controller response from the change in cooling water inlet temperature and reactor temperature versus time

Code 1: Feedback and Cascade P1

!Variable Declarations and Data

!*****

!William Rawson

!CHE 341

Parameter Rg=8.3145 As Real !Universal Gas Constant, J/(mol K)

!*****Declare Variables*****

!Declare Feed Variables

Global Q, CaFeed, CbFeed, CcFeed, CdFeed, TFeed, Qcw As Real

Q = 0.454249 !m3/min

Qcw = 97 !gpm

CaFeed = 240.173/60 !kmol/min

CbFeed = 232.41/60 !kmol/min

CcFeed = 0.00 !kmol/min

CdFeed = 12.615/60 !kmol/min

TFeed = 25. !C

!Declare Reactor/Jacket Variables

Global Vol, UA, Tc, Tr, Tcin, Vjack As Real

Vol = 10 !m3

UA = 928 !kJ/min-K

Tcin = 25 !C

Tr = 273.15 + 70. !K

Vjack = 0.808 !m3

!Declare Physical Property Variables

Global rhof, rhop, rhofM, Cpf, Cpp, Cp, DHR1, rhoWM, Cpw As Real

rhofM = 17.798 !kmol/m3

rhoWM = 55.18 !kmol/m3

Cpf = 100.032 !kJ/kmol-K

Cpp = 2.548246 !kJ/kg-K

Cpw = 74.22 !kJ/kmol-K

DHR1 = -29.0697e3 !kJ/kmol

!Declare Reaction Property Variables

Global k0fw, k0r, Er, Efw, cv, sqrtP As Real

k0fw = 1.9E8 !m3/kmol-s

k0r = 5.0e7 !m3/kmol-s

Er = 59500 !J/mol

Efw = 59500 !J/mol

cv = 36

sqrtP = sqrt(7.26) !psi^0.5

@Initial Conditions to find steady state conditions

!U(1)=rhofM*0.5

!U(2)=0.0

!U(3)=0.0

!U(4)=rhofM*0.5

!U(5)= 50 !C

!U(6)= 25 !C

!Steady State Conditions (U(1)-U(6)): 3.16272E+00 2.87789E+00 5.64937E+00 6.11222E+00 7.89546E+01 4.55878E+01

U(1) = 3.16272E+00

U(2) = 2.87789E+00

U(3) = 5.64937E+00

U(4) = 6.11222E+00

U(5) = 7.89546E+01

U(6) = 4.55878E+01

@Model Equations

Dim Ca, Cb, Cc, Cd, Temp, Tempw As Real

Ca = U(1)

Cb = U(2)

Cc = U(3)


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Cd = U(4)
Temp = U(5) + 273.15
Tempw = U(6) !C

Dim kf,kr,R1,R2 As Real
kf = k0fw * exp (-Efw / ( Rg * Temp ))
kr = k0r * exp (-Er / ( Rg * Temp ))
R1 = kf * Ca * Cb - kr * Cc * Cd
R1 = R1 * 60

If(iRange=1)cv = 60.4
If(iRange=2)cv = 21.7
If(iRange=3)cv = 36
F(1)= CaFeed-Q*Ca-R1*Vol
F(2)= CbFeed-Q*Cb-R1*Vol
F(3)= CcFeed-Q*Cc+R1*Vol
F(4)= CdFeed-Q*Cd+R1*Vol
F(5)= Q*rhofM*Cpf*(TFeed-U(5))+(-DHR1*R1*Vol)-UA*(U(5)-U(6))
F(6)=(cv*sqrtP*0.003786*rhoWM*Cpw)*(Tcin-U(6))-UA*(U(6)-U(5))

@Coefficient Matrix
E(1)= Vol
E(2)= Vol
E(3)= Vol
E(4)= Vol
E(5)= rhofM*Cpf*Vol
E(6)=rhoWM*Vjack*Cpw

@Solver Options
Ncq=6
Npts=100
Tend=1000
Tstop=200;400;600;800
Param(28)=1
VariableNames=Time(min);Ca;Cb;Cc;Cd, Temperature Reactor (C);Temperature Cooling Jacket (C)

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Code 2: Cascade P2

!Variable Declarations and Data

!*****

!William Rawson

!CHE 341

Parameter Rg=8.3145 As Real !Universal Gas Constant, J/(mol K)

!*****Declare Variables*****

!Declare Feed Variables

Global Q, CaFeed, CbFeed, CcFeed, CdFeed, TFeed, Qcw As Real

Q = 0.454249 !m3/min

Qcw = 97 !gpm

CaFeed = 240.173/60!kmol/min

CbFeed = 232.41/60!kmol/min

CcFeed = 0.00 !kmol/min

CdFeed = 12.615/60 !kmol/min

TFeed = 25. !C

!Declare Reactor/Jacket Variables

Global Vol, UA, Tc, Tr, Tcin, Vjack As Real

Vol = 10 !m3

UA = 928 !kJ/min-K

Tcin = 25 !C

Tr = 273.15 + 70. !K

$$V_{\text{jack}} = 0.808 \text{ m}^3$$

!Declare Physical Property Variables
 Global rhoF,rhoP,rhoM,CpF,CpP,Cp,DHR1,rhoWM,Cpw As Real
 rhoM = 17.798 !kmol/m3
 rhoWM = 55.18 !kmol/m3
 CpF = 100.032 !kJ/kmol-K
 CpP = 2.548246 !kJ/kg-K
 Cpw = 74.22 !kJ/kmol-K
 DHR1 = -29.0697e3 !kJ/kmol

!Declare Reaction Property Variables
 Global k0fW,k0r,Er,EfW,cv,sqrtP As Real
 k0fW = 1.9E8 !m3/kmol-s
 k0r = 5.0e7 !m3/kmol-s
 Er = 59500 !J/mol
 EfW = 59500 !J/mol
 cv = 36
 sqrtP = sqrt(7.26) !psi^0.5

@Initial Conditions to find steady state conditions

!U(1)=rhoM*0.5
 !U(2)=0.0
 !U(3)=0.0
 !U(4)=rhoM*0.5
 !U(5)= 50 !C
 !U(6)= 25 !C

!Steady State Conditions (U(1)-U(6)): 3.16272E+00 2.87789E+00 5.64937E+00 6.11222E+00 7.89546E+01 4.55878E+01

U(1)= 3.16272E+00
 U(2)= 2.87789E+00
 U(3)= 5.64937E+00
 U(4)= 6.11222E+00
 U(5)= 7.89546E+01
 !U(6)= 4.55878E+01

@Model Equations

Dim Ca, Cb, Cc, Cd, Temp, Tempw As Real
 Ca = U(1)
 Cb = U(2)
 Cc = U(3)
 Cd = U(4)
 Temp = U(5)+ 273.15
 Tempw = 45.59

Dim kf,kr,R1,R2 As Real

kf = k0fW * exp (-EfW / (Rg * Temp))
 kr = k0r * exp (-Er/ (Rg * Temp))
 R1 = kf * Ca * Cb - kr * Cc * Cd
 R1 = R1 * 60

If(iRange=1)Tempw = 50.59

If(iRange=2)Tempw = 40.59

If(iRange=3)Tempw = 45.59

F(1)= CaFeed-Q*Ca-R1*Vol

F(2)= CbFeed-Q*Cb-R1*Vol

F(3)= CcFeed-Q*Cc+R1*Vol

F(4)= CdFeed-Q*Cd+R1*Vol

F(5)= Q*rhoM*CpF*(TFeed-U(5))+(-DHR1*R1*Vol)-UA*(U(5)-Tempw)

!F(6)=(cv*sqrtP*0.003786*rhoWM*Cpw)*(Tcin-U(6))-UA*(U(6)-U(5))

@Coefficient Matrix

E(1)= Vol

E(2)= Vol

E(3)= Vol

E(4)= Vol

E(5)= rhoM*CpF*Vol

!E(6)=rhoWM*Vjack*Cpw

@Solver Options

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Neq=5
Npts=100
Tend=1000
Tstop=200;400;600;800
Param(28)=1
VariableNames=Time(min);Ca;Cb;Cc;Cd; Temperature Reactor (C);Temperature Cooling Jacket (C)

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Code 3: Feed forward

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!Variable Declarations and Data
!*****
!William Rawson
!CHE 341

Parameter Rg=8.3145 As Real !Universal Gas Constant, J/(mol K)

!*****Declare Variables*****

!Declare Feed Variables

Global Q, CaFeed, CbFeed, CcFeed, CdFeed, TFeed, Qcw As Real
Q = 0.454249 !m3/min
Qcw = 97 !gpm
CaFeed = 240.173/60!kmol/min
CbFeed = 232.41/60!kmol/min
CcFeed = 0.00 !kmol/min
CdFeed = 12.615/60 !kmol/min
TFeed = 25. !C

!Declare Reactor/Jacket Variables
Global Vol, UA, Tc, Tr, Tcin, Vjack As Real
Vol = 10 !m3
UA = 928 !kJ/min-K
Tcin = 25 !C
Tr = 273.15 + 70. !K
Vjack = 0.808 !m3

!Declare Physical Property Variables
Global rhof, rhop, rhofM, Cpf, Cpp, Cp, DHR1, rhoWM, Cpw As Real
rhofM = 17.798 !kmol/m3
rhoWM = 55.18 !kmol/m3
Cpf = 100.032 !kJ/kmol-K
Cpp = 2.548246 !kJ/kg-K
Cpw = 74.22 !kJ/kmol-K
DHR1 = -29.0697e3 !kJ/kmol

!Declare Reaction Property Variables
Global k0fw, k0r, Er, Efw, cv, sqrtP As Real
k0fw = 1.9E8 !m3/kmol-s
k0r = 5.0e7 !m3/kmol-s
Er = 59500 !J/mol
Efw = 59500 !J/mol
cv = 36
sqrtP = sqrt(7.26) !psi^0.5

@Initial Conditions to find steady state conditions
!U(1)=rhofM*0.5
!U(2)=0.0
!U(3)=0.0
!U(4)=rhofM*0.5
!U(5)= 50 !C

```

```

!U(6)= 25 !C

!Steady State Conditions (U(1)-U(6)): 3.16272E+00 2.87789E+00 5.64937E+00 6.11222E+00 7.89546E+01 4.55878E+01
U(1)= 3.16272E+00
U(2)= 2.87789E+00
U(3)= 5.64937E+00
U(4)= 6.11222E+00
U(5)= 7.89546E+01
U(6)= 4.55878E+01

@Model Equations
Dim Ca, Cb, Cc, Cd, Temp, Tempw As Real
Ca = U(1)
Cb = U(2)
Cc = U(3)
Cd = U(4)
Temp = U(5)+ 273.15
Tempw = U(6) !C

Dim kf,kr,R1,R2 As Real
kf= k0fw * exp (-Efw /( Rg * Temp ))
kr= k0r * exp (-Er/( Rg * Temp ))
R1 = kf * Ca * Cb - kr * Cc * Cd
R1 = R1 * 60

If(iRange=1)Tcin =30
If(iRange=2)Tcin = 20
If(iRange=3)Tcin = 25
F(1)= CaFeed-Q*Ca-R1*Vol
F(2)= CbFeed-Q*Cb-R1*Vol
F(3)= CcFeed-Q*Cc+R1*Vol
F(4)= CdFeed-Q*Cd+R1*Vol
F(5)= Q*rhofM*Cpf*(TFeed-U(5))+(-DHR1*R1*Vol)-UA*(U(5)-U(6))
F(6)=(cv*sqrtP*0.003786*rhoWM*Cpw)*(Tcin-U(6))-UA*(U(6)-U(5))

@Coefficient Matrix
E(1)= Vol
E(2)= Vol
E(3)= Vol
E(4)= Vol
E(5)= rhofM*Cpf*Vol
E(6)=rhoWM*Vjack*Cpw

@Solver Options
Neq=6
Npts=100
Tend=1000
Tstop=200;400;600;800
Param(28)=1
VariableNames=Time(min);Ca;Cb;Cc;Cd; Temperature Reactor (C);Temperature Cooling Jacket (C)

```